

One Interval, Two Cutoffs

AP Statistics · Unit 4 · Topic 4.3 · TEACHER KEY

CED TETHER

- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **Skill 4.D** — Justify a claim based on a confidence interval.
- **Skill 4.A** — Make an appropriate claim or draw an appropriate conclusion.
- **EU UNC-4** — An interval of values should be used to estimate parameters, in order to account for uncertainty.
- **LO UNC-4.S** — Interpret a confidence interval for a population mean, including the mean difference between values in matched pairs. [Skill 4.B]
- **EK UNC-4.S.1** — A confidence interval for a population mean either contains the population mean or it does not, because each interval is based on data from a random sample, which varies from sample to sample.
- **EK UNC-4.S.2** — We are C% confident that the confidence interval for a population mean captures the population mean.
- **EK UNC-4.S.3** — An interpretation of a confidence interval for a population mean includes a reference to the sample taken and details about the population it represents.
- **LO UNC-4.T** — Justify a claim based on a confidence interval for a population mean, including the mean difference between values in matched pairs. [Skill 4.D]
- **EK UNC-4.T.1** — A confidence interval for a population mean provides an interval of values that may provide sufficient evidence to support a particular claim in context.
- **LO UNC-4.U** — Identify the relationships between sample size, width of a confidence interval, confidence level, and margin of error for a population mean. [Skill 4.A]
- **EK UNC-4.U.1** — When all other things remain the same, the width of a confidence interval for a population mean tends to decrease as the sample size increases.
- **EK UNC-4.U.2** — For a single mean, the width of the interval is proportional to $1/\sqrt{n}$.
- **EK UNC-4.U.3** — For a given sample, the width of the confidence interval for a population mean increases as the confidence level increases.

Essential question: How can one confidence interval support a bounded claim while leaving a stronger claim unresolved?

DOK-3 skill: 4.D · **FRQ pattern:** mean-interval-two-cutoffs-and-scope

DOK rationale: Part (c) coordinates both endpoints, two cutoffs, random-sample scope, and the mean-versus-individual distinction to adjudicate two interpretations and trace an advertising consequence.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which specific number or study detail supports your choice?
- What claim would go beyond the evidence, and why?

[WARN] LOOK FOR

- Choosing a speaker without a reason.
- Copying numbers without explaining what they support.

SCORING PART (c) — E / P / I

Expected elements

- **judges-45** (required) — Uses the upper endpoint below 45 to support the mean-below-45 claim
- **judges-42** (required) — Uses 42 inside the interval to reject support for the stronger claim
- **mean-not-individuals** (required) — Explains that the interval estimates a mean rather than containing 95% of transactions

E	Uses both cutoff locations correctly and distinguishes the mean from individual transaction times.
P	Shows substantive valid reasoning for at least one required element, but does not meet all E requirements. Credit correct reasoning even when another claim is wrong.
I	Shows no substantive valid reasoning for any required element; a name, unsupported choice, or copied number alone does not count.

[STAR] THE PROBLEM — annotated

Suppose a library wants the mean setup time for all 800 kiosk transactions in a month below 45 seconds and hopes to advertise a mean below 42 seconds. An SRS of 25 has the shown summary. A 95% one-sample t -interval is (39.923, 44.877) seconds.

Mina: “I would compare each cutoff with the full interval and keep the conclusion about the population mean.”

Gabe: “Because the interval contains times below both cutoffs, I think it supports both claims; I also read 95% as the share of transaction times inside it.”

Kiosk sample

Summary	Value
Population	800
n	25
Mean (s)	42.4
SD (s)	6.0
Shape	Symmetric; no outliers

FIRST TAKE — one sentence before you work the parts

Does the range 39.923 to 44.877 make both time claims believable? Give one reason.

Key: Ungraded commitment; accept any evidence-based first impression.

[BOOK] LET THE ENTIRE INTERVAL GOVERN THE CLAIM (keep on the page)

A $C\%$ confidence interval estimates a population mean, not the spread of individual observations. We are $C\%$ confident that the interval captures the population mean represented by the random sample; the fixed interval either contains that mean or it does not. A value inside the interval remains plausible. To support $\mu < c$, the entire interval must lie below c ; an endpoint merely falling below c is not enough. With other features held fixed, a larger sample tends to narrow a mean interval, while a higher confidence level widens it.

DOK 1 (a) Locate 42 and 45 relative to the interval.

Key: The value 42 is inside (39.923, 44.877); 45 is above its upper endpoint.

DOK 2 (b) Interpret the supplied 95% interval in one sentence about the mean setup time for all 800 transactions.

Key: We are 95% confident that the mean setup time for all 800 kiosk transactions that month is between 39.923 and 44.877 seconds.

DOK 3 (c) In 3–4 sentences, decide whose approach is justified. Use the interval to judge each cutoff, then explain what Gabe gets wrong about individual transaction times.

Key: Mina is justified: the entire interval is below 45, supporting a population mean below 45 seconds. Because 42 is inside, means above 42 remain plausible, so a mean below 42 is not established. The interval estimates one population mean; it does not describe the middle 95% of individual transaction times. The library should therefore avoid advertising a mean below 42 or a guarantee about individual times.